



- 3 (a)** Use standard results from the list of formulae (MF19) to show that

$$\sum_{r=1}^n (r+1)(r+2)(r+3) = \frac{1}{4}n(n^3 + bn^2 + cn + d),$$

where b, c and d are integers to be determined.

[3]

[illegible]



- (b)** Express $\frac{2}{(r+1)(r+2)(r+3)}$ in partial fractions and hence use the method of differences to find

$$\sum_{r=1}^n \frac{2}{(r+1)(r+2)(r+3)}. \quad [5]$$

This image shows a full page of a worksheet designed for handwriting practice. It features approximately 20 horizontal dashed lines spaced evenly across the page, providing a guide for letter height and placement. The background is plain white, and there are no other markings or text present.

- (c) State the value of $\sum_{r=1}^{\infty} \frac{2}{(r+1)(r+2)(r+3)}$. [1]

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